ИДЗ N 1 I вар

1. $\frac{2}{3}xyy' = \sqrt{x^6 - y^4} + y^2$, $x > 0$, $y > 0$

• $y = z^m \Rightarrow y' = mz^{m-1}z'$ - замена

Ур-ие принимает вид:

$\frac{2}{3}xz^m mz^{m-1}z' = \sqrt{x^6 - z^{4m}} + z^{2m}$;

Для однородности необходимо:

$m + 1 + m - 1 = \sqrt{6 - 4m} + 2m \Rightarrow 6 - 4m = 0 \Rightarrow m = \frac{3}{2}$

Тогда: $\frac{2}{3}xz^{\frac{3}{2}}\frac{3}{2}z^{\frac{1}{2}}z' = \sqrt{x^6 - z^6} + z^3 \Rightarrow$

$\Rightarrow xz^2z' = \sqrt{x^6 - z^6} + z^3$;

• Произведем замену $z = t(x)x \Rightarrow$ $t = \frac{z}{x}$

$\Rightarrow dz = tdt + tdx \Rightarrow$

$\Rightarrow x^3t^2(x)\frac{dz}{dx} = \sqrt{x^6 - t^6(x)x^6} + t^3(x)x^3 \;|:x^3$

$t^2dz = (\sqrt{1 - t^6} + t^3)dx$

$t^2xdt + t^3dx = (\sqrt{1 - t^6} + t^3)dx \;|: x\sqrt{1 - t^6}$

$\frac{t^2}{\sqrt{1 - t^6}}dt = \frac{dx}{x} \Rightarrow \frac{1}{3}\int \frac{dt^3}{\sqrt{1 - t^6}} = \ln x$; $|t| < 1$, $t = 1$ - решение

$\frac{\arcsin t}{3} - \ln x = C \Rightarrow \arcsin\frac{z}{x} - 3\ln x = C \Rightarrow$

$\Rightarrow \arcsin\frac{y^3}{x^3} - 3\ln x = C$

2. $y' = \frac{y}{x} + x,\ x > 0$

• $y' = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y}{x} \Rightarrow \frac{dy}{y} = \frac{dx}{x} \Rightarrow \ln|y| - \ln x = \tilde{C},\ C \in \mathbb{R}$

$\ln|y| = \ln x + \ln \tilde{C},\ \tilde{C} > 0 \Rightarrow y = Cx,\ C \in \mathbb{R}$

• Т.к. $C = C(x)$, то $y' = C'x + \underbrace{x'}_{=1}C$

Подставим в исходное ур-е:

$C'x + \underbrace{x'}_{=1}C = C + x$

$C'x + C = C + x$

$\frac{dC}{dx} = 1 \Rightarrow C = x + C^*$

Подставим в у: $y = x^2 + C^*x$.

3. $3y' + y^2 + \frac{2}{x^2} = 0,\ x > 0$

$y(x) = \frac{a}{x};\ -\frac{3a}{x^2} + \frac{a^2}{x^2} + \frac{2}{x^2} = 0 \Rightarrow a^2 - 3a + 2 = 0 \Rightarrow (a-1)(a-2) = 0$

$\Rightarrow \left[\begin{array}{l} a = 1 \\ a = 2 \end{array}\right.$ .. частное решение: $-\frac{3}{x^2} + \frac{1}{x^2} + \frac{2}{x^2} = 0$

$y = z + \frac{1}{x};\ y' = z' - \frac{1}{x^2} \Rightarrow$

$\Rightarrow 3z' - \frac{3}{x^2} + z^2 + \frac{2z}{x} + \frac{1}{x^2} + \frac{2}{x^2} = 0 \Rightarrow$

$\Rightarrow 3z' + z^2 + \frac{2z}{x} = 0$ – уравнение Бернулли

$3z' + z^2 + \frac{2z}{x} = 0 \ |: 3z^2 \Rightarrow \frac{z'}{z^2} + \frac{1}{3} + \frac{2}{3zx} = 0$

Сделаем замену $\rho = z^{-1};\ \rho' = -z^{-2}z'$

$+z^2 z' + \frac{2z^{-1}}{3x} + \frac{1}{3} = -\rho' + \frac{2\rho}{3x} + \frac{1}{3} = 0 \Rightarrow \rho' = \frac{2\rho}{3x} + \frac{1}{3}$

$z = \frac{1}{\rho}$

$y = \frac{1}{\rho} + \frac{1}{x} \Rightarrow$

$\Rightarrow \frac{yx - 1}{x} = \frac{1}{\rho}$

и решим методом интегрирующего множителя:

$$p' = \frac{dp}{dx} = \frac{2p}{3x} + \frac{1}{3} \Rightarrow dp + \left(-\frac{2p}{3x} - \frac{1}{3}\right)dx = 0 \Rightarrow$$

$$\Rightarrow \underbrace{\left(-\frac{2p}{3x} - \frac{1}{3}\right)}_{P(x,p)}dx + \underbrace{1}_{Q(x,p)}\cdot dp = 0 \quad (1)$$

$$\mu(x) = e^{\int\left(-\frac{2dx}{3x}\right)} = e^{-\frac{2}{3}\ln x} = x^{-\frac{2}{3}}$$

Домножим ур-е (1) на $x^{-\frac{2}{3}}$:

$$\underbrace{\left(-\frac{2p}{3x^{5/3}} - \frac{1}{3x^{2/3}}\right)}_{P(x,p)}dx + \underbrace{\frac{dp}{x^{2/3}}}_{Q(x,p)} = 0 \quad \text{– ур-е в полных дифф-х}$$

$$\frac{\partial u}{\partial x} = P(x,\ ); \quad \frac{\partial u}{\partial p} = Q(x,p)$$

$$du = \left(-\frac{2p}{3x^{5/3}} - \frac{1}{3x^{2/3}}\right)dx + \frac{dp}{x^{2/3}}$$

$$u(x,p) = \int\left(\frac{-2p}{3x^{5/3}} - \frac{1}{3x^{2/3}}\right)dx = \frac{p}{x^{2/3}} + \frac{1}{x^{1/3}} + \tilde{\tilde{C}}(p)$$

$$u_p'(x,p) = \frac{1}{x^{2/3}} + \tilde{\tilde{C}}'(\ ) = Q(x,p) \Rightarrow \tilde{\tilde{C}}'(\ ) = 0 \Rightarrow$$

$$\Rightarrow u = \frac{p}{x^{2/3}} + \frac{1}{x^{1/3}} + \tilde{C} \Rightarrow \frac{p}{x^{2/3}} + \frac{1}{x^{1/3}} = C = C_1 - \tilde{C} \Rightarrow$$

C_1 – ?

$$\Rightarrow \frac{x^{1/3}}{yx-1} + \frac{1}{x^{1/3}} = C = F(x,y) \qquad \left(p = \frac{x}{yx-1}\right)$$

! $z = 0$ – решение $\Rightarrow y = \frac{1}{x}$ – решение тоже